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Deveswar

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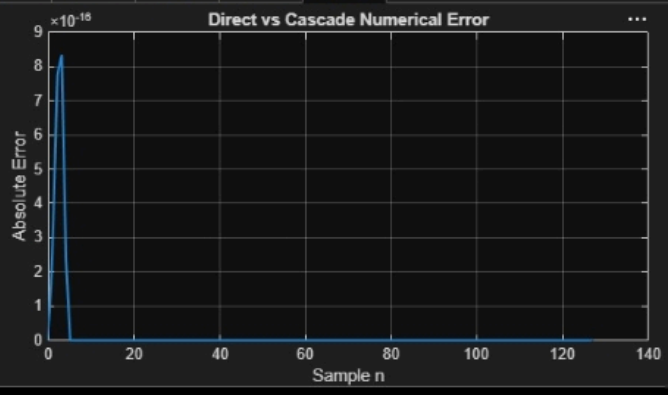
ip.m × | exp3dsp.m × | exp4dsp.m × | exp5dsp.m × | exp6dsp.m × | exp7dsp.m × | exp8dsp.m ×

/MATLAB Drive/exp8dsp.m

```
213
214     title('Direct vs Cascade Numerical Error');
215
216     %% ========================================================
217     % FINAL RESULT
218     % ========================================================
219
220     fprintf('\n=========================================\n');
221     fprintf('FINAL RESULT\n');
222     fprintf('=========================================\n');
223
224     fprintf('1. FIR filter realized using Direct structure.\n');
225     fprintf('2. FIR filter realized using Cascade structure.\n');
226     fprintf('3. Coefficients are %s.\n',symmetry);
227     fprintf('4. Linear-phase group delay = %.2f samples.\n',D);
228     fprintf('5. Maximum realization error = %.3e.\n',err);
229     fprintf('6. Percentage deviation = %.3e %%.\n',dev);
230     fprintf('7. Direct and Cascade responses are equivalent.\n');
```

Figures

Figure 1 × | Figure 2 × | Figure 3 × | Figure 4 × | Figure 5 ×



Command Window

```
1. FIR filter realized using Direct structure.
2. FIR filter realized using Cascade structure.
3. Coefficients are Symmetric.
4. Linear-phase group delay = 2.00 samples.
5. Maximum realization error = 8.327e-16.
6. Percentage deviation = 2.220e-13 %.
7. Direct and Cascade responses are equivalent.
```

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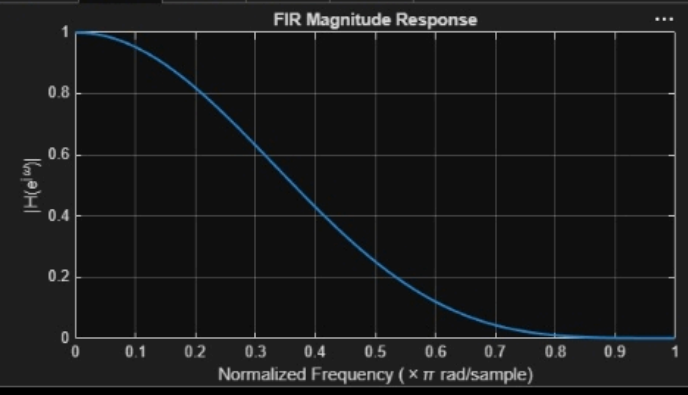
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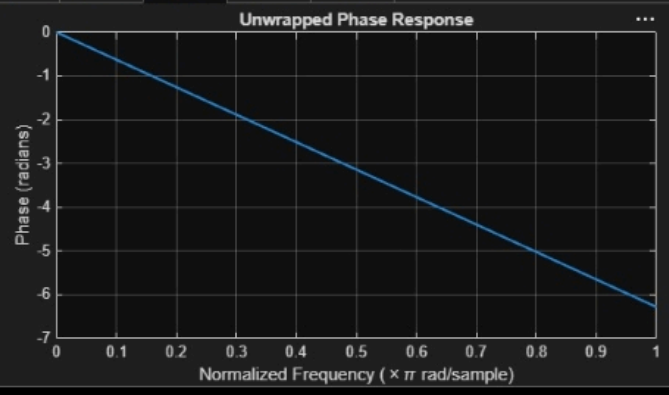
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```
1   clc; clear; close all;
2
3   %% FIR FILTER PARAMETERS
4   b = [0.0625 0.2500 0.3750 0.2500 0.0625];
5   Fs = 1000;
6
7   N = 128;
8
9   x = [1 zeros(1,N-1)];
10  n = 0:N-1;
11
12  %% 1. DIRECT REALIZATION
13  yd = filter(b,1,x);
14
15  %% 2. CASCADE REALIZATION
16  z = roots(b);
17  g = b(1);
18  yc = g*x;
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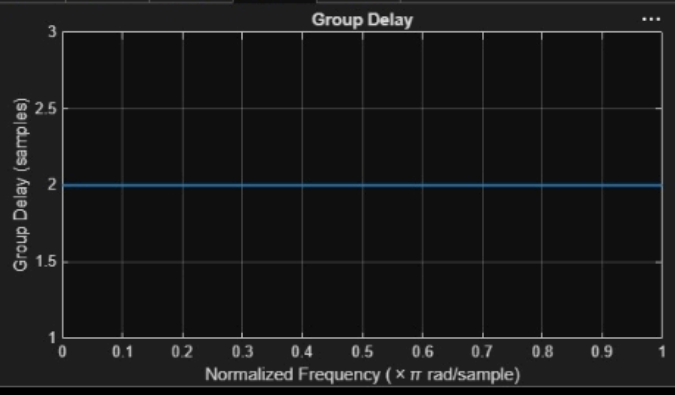
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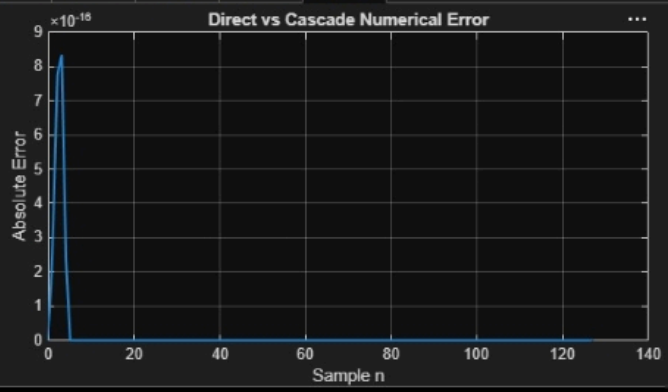
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